

Mahmoud Adas

Software Engineer - Egypt

☎ (+20) 1019-42-6003 | ✉ mahmoud.othman.adas@gmail.com | 🌀 mido3ds | 🌐 mahmoud-adas

About Me

C++ software engineer with 4 years building desktop dental CAD applications at Atomica.ai. Led the general-cad team and built internal tooling (SQLite code generation, library editor, CI pipelines). Strong Qt/QML skills, with side interests in emulation, network protocols, and flight simulation.

Education

Cairo University, Faculty of Engineering | Egypt
B.S. in Computer Engineering

Sep 2015 - Jul 2021

Tools

- C++, C, Go, Python, Java, Rust
- Qt/QML, React, Spring, Android
- SQLite, Docker, Linux

Experience

C++ Software Engineer

Jun 2021 - Mar 2022, Jun 2023 - Now

Atomica.ai (360Imaging) - Giza, Egypt

- Led the general-cad dental application team
- Maintained the frontend of dental implant planning and teeth segmentation applications, handling cross-team feature integration and bug fixes
- Developed an experimental Qt/QML launcher using libgit for file synchronization and a custom version-tracking file format — some components were later adopted by the main product suite
- Built an ImGui/SQLite Library Editor enabling product managers to update dental catalog data (implants, abutments, materials) and release to AWS — eliminating app releases for data changes
- Contributed to the company's core C++ libraries (mn utility library, ndcore SDK) used across all product applications
- Built internal C++ tooling including ldb2cpp (SQLite-to-C++ code generator), Library Editor (data authoring tool), and CI/test infrastructure — designed to be the standard approach for database workflows
- Created a SQLite-to-C++ code generator (ldb2cpp) producing type-safe database bindings, cutting boilerplate across multiple internal projects

Software Engineering Intern · Cert

Jul - Sep 2020

Solectra - El Gouma, Egypt

- Maintained an **embedded system** that collects and analyses data from electric vehicles to report trips, total revenue, and energy
- Fixed embedded system bugs and improved EV trip/revenue prediction accuracy through data analysis and additional test coverage

Projects

360Launcher

Jan - Mar 2022

Qt/QML/C++ app to install/update/launch main products and integrate with them

Library Editor

Sep - Dec 2021

C++ (+ImGui) internal tool to modify/upload versioned metadata and information. Helps product managers update domain-specific logic without new app release.

C4IAN (Graduation Project) · Repo

Feb - Jul 2021

- Tactical Mobile ad-hoc network. Supports video/audio streaming and messaging
- ZHLS+ZID unicast, ODMRP multicast, and geo-based broadcast
- **Golang** router that integrates with the Linux network stack

OpenYSF · Repo

Nov 2021 - Present

A ysflight flight simulator clone, for learning and fun. Backward compatible with ysflight assets.

Apex · Backend · Frontend · Android · Testing

Jan - Mar 2019

- Reddit clone, for web and android. Took role of **Devops** in team of 18
- Used Travis-CI for CI/CD, **GoogleCloud**, **Docker**, DockerCompose, and Datadog for monitoring

NesEmu · Repo

2019 - Present

NES console **emulator** in C++. Hobby project.

Koos · Repo · Rust Rewrite

Jun 2019

- Decentralized social media app, built on Fluence. AssemblyScript/Rust for server, React for frontend
- Won a Bitcoin bounty for the best app that uses Fluence's technology

Match Reservation System · Repo

Jan - Feb 2021

Web application for Egyptian Premier League for seats reservation. Used **MongoDB**, **Nodejs** and React